

ACADEMIC EXPERIENCE

University of Pennsylvania

Postdoctoral Researcher

– Hosted by Prof. Weijie Su & Prof. Qi Long

Philadelphia, PA, USA

2023 - present

EDUCATION

The Chinese University of Hong Kong, Shenzhen

Ph.D. in Computer and Information Engineering

– Advisor: Prof. Zhi-Quan (Tom) Luo

– Thesis: Understanding Adversarially Robust Generalization: A Learning Theory Perspective

Shenzhen, China

2017 - 2023

The Chinese University of Hong Kong

M.S. in Mathematics

Hong Kong SAR, China

2014 - 2015

Sun Yat-sen University

B.Econ, Mathematics Honors Program

– Honors program with a dual curriculum in finance and mathematics

Guangzhou, China

2010 - 2014

RESEARCH INTEREST

I am broadly interested in **learning theory, statistics and optimization**, with a focus on establishing foundations for **responsible** and **trustworthy** machine learning models. My recent research focuses on **mathematical and statistical topics in large language models (LLMs)**.

- **Theoretical Foundations for LLM Alignment:** Analyzing algorithmic biases in preference alignment methods [J1, P2], exploring game-theoretic approaches to alignment [J2], and investigating alignment from the perspective of social choice theory [P1, P3].
- **LLM Training and Fine-Tuning:** Establishing theoretical foundations for fine-tuning and post-training algorithms [C2, C3, C4, W1]. Addressing calibration issues [C1] in LLMs.
- **Adversarial Robustness:** Developing theoretical frameworks for understanding adversarial examples [J4, W2], robust overfitting [C8, C6], and adversarially robust generalization [J3, C5, C7].
- **Classical Learning Theory:** Studying optimization (convergence and stability of non-convex, non-smooth problems [C8, C6, W4]) and generalization (Rademacher complexity [J3, C5], PAC-Bayes [C7, W3]).

PUBLICATIONS

(α - β) denotes alphabetical order, * denotes equal contribution, and [†] denotes corresponding authors.

Journal Articles (Published or Under Revision)

- [J1] On the Algorithmic Bias of Aligning Large Language Models with RLHF: Preference Collapse and Matching Regularization.
Jiancong Xiao, Ziniu Li, Xingyu Xie, Emily Getzen, Cong Fang, Qi Long, and Weijie J. Su.
Journal of the American Statistical Association (JASA), 1–21, 2025.
- [J2] Statistical Impossibility and Possibility of Aligning LLMs with Human Preferences: From Condorcet Paradox to Nash Equilibrium.
(α - β) Kaizhao Liu, Qi Long, Zhekun Shi, Weijie J. Su, and **Jiancong Xiao**.
Annals of Statistics (AoS), major revision, 2025+.

- [J3] Adversarial Rademacher Complexity of Deep Neural Networks.
Jiancong Xiao, Yanbo Fan, Ruoyu Sun, and Zhi-Quan Luo.
Journal of Machine Learning Research (JMLR), invited revision submitted, 2025+.
- [J4] Understanding Adversarial Robustness Against On-Manifold Adversarial Examples.
Jiancong Xiao, Liusha Yang, Yanbo Fan, Jue Wang, and Zhi-Quan Luo.
Pattern Recognition, 159, 111071, 2025.

Conference Papers

- [C1] Restoring Calibration for Aligned Large Language Models: A Calibration-Aware Fine-Tuning Approach.
Jiancong Xiao^{*}, Bojian Hou^{*}, Zhanliang Wang^{*}, Ruochen Jin, Qi Long, Weijie J. Su, and Li Shen.
The 42nd International Conference on Machine Learning (ICML), 68364–68390, 2025.
- [C2] Fine-Tuning Attention Modules Only: Enhancing Weight Disentanglement in Task Arithmetic.
Ruochen Jin^{*}, Bojian Hou^{*}, **Jiancong Xiao**^{*}, Weijie J. Su, and Li Shen.
The Thirteenth International Conference on Learning Representations (ICLR), 2025.
- [C3] Preserving Diversity in Supervised Fine-tuning of Large Language Models.
Ziniu Li, Congliang Chen, Tian Xu, Zeyu Qin, **Jiancong Xiao**, Zhi-Quan Luo, and Ruoyu Sun.
The Thirteenth International Conference on Learning Representations (ICLR), 2025.
- [C4] Magnetic Mirror Descent Self-play Preference Optimization.
Mingzhi Wang, Chengdong Ma, Qizhi Chen, Linjian Meng, Yang Han, **Jiancong Xiao**, Zhaowei Zhang, Jing Huo, Weijie J. Su, and Yaodong Yang.
The Thirteenth International Conference on Learning Representations (ICLR), 2025.
- [C5] Bridging the Gap: Rademacher Complexity in Robust and Standard Generalization.
Jiancong Xiao, Ruoyu Sun, Qi Long, and Weijie J. Su.
The Thirty Seventh Conference on Learning Theory (COLT), 5074–5075, 2024.
- [C6] Uniformly Stable Algorithms for Adversarial Training and Beyond.
Jiancong Xiao[†], Jiawei Zhang[†], Zhi-Quan Luo, and Asuman E. Ozdaglar.
The 41st International Conference on Machine Learning (ICML), 54319–54340, 2024.
- [C7] PAC-Bayesian Spectrally-Normalized Bounds for Adversarially Robust Generalization.
Jiancong Xiao, Ruoyu Sun, and Zhi-Quan Luo.
Advances in Neural Information Processing Systems (NeurIPS), 36, 36305–36323, 2023.
- [C8] Stability Analysis and Generalization Bounds of Adversarial Training.
Jiancong Xiao, Yanbo Fan, Ruoyu Sun, Jue Wang, and Zhi-Quan Luo.
Advances in Neural Information Processing Systems (NeurIPS Spotlight), 35, 15446–15459, 2022.

Preprints & Working Papers

- [P1] Theoretical Tensions in RLHF: Reconciling Empirical Success with Inconsistencies in Social Choice Theory.
Jiancong Xiao, Zhekun Shi, Kaizhao Liu, Qi Long, and Weijie J. Su.
Submitted, under review, 2025.
- [P2] A Fine-Grained Analysis of Pure Semantic Preference Alignment in Large Language Models.
Ruochen Jin^{*}, **Jiancong Xiao**^{*}, Nikhil Ruia, Qi Long, and Weijie J. Su.
Submitted, under review, 2025.
- [P3] Fundamental Limits of Game-Theoretic LLM Alignment: Smith Consistency and Preference Matching.
Zhekun Shi, Kaizhao Liu, Qi Long, Weijie J. Su, and **Jiancong Xiao**[†].
Submitted, under review, 2025.

Workshop Papers

- [W1] Entropic Distribution Matching in Supervised Fine-tuning of LLMs: Less Overfitting and Better Diversity.
Ziniu Li, Congliang Chen, Tian Xu, Zeyu Qin, **Jiancong Xiao**, Ruoyu Sun, and Zhi-Quan Luo.
NeurIPS 2024 Workshop on Fine-Tuning in Modern Machine Learning: Principles and Scalability (Best Paper Runner-up), 2024.
- [W2] Improving Adversarial Training for Multiple Perturbations through the Lens of Uniform Stability.
Jiancong Xiao, Zeyu Qin, Yanbo Fan, Baoyuan Wu, Jue Wang, and Zhi-Quan Luo.
ICML Workshop on New Frontiers in Adversarial Machine Learning, 2023.

- [W3] PAC-Bayesian Adversarially Robust Generalization Bounds for Deep Neural Networks.
Jiancong Xiao, Ruoyu Sun, and Zhi-Quan Luo.
ICML Workshop on New Frontiers in Adversarial Machine Learning, 2023.
- [W4] Smoothed-SGDmax: A Stability-Inspired Algorithm to Improve Adversarial Generalization.
Jiancong Xiao, Jiawei Zhang, Zhi-Quan Luo, and Asuman E. Ozdaglar.
NeurIPS ML Safety Workshop, 2022.

GRANTS & FUNDING

- Contributor to NIH RO1 Grant Proposal, co-written with Prof. Weijie Su & Prof. Qi Long (2024).
 - Assisted in proposal writing, structuring key research components.

TALKS AND PRESENTATION

Invited Talks

- LLM Training and Coding for Statistical Learning and Data Science American Statistical Association, 2025.11
- Statistical Limits of LLM Alignment Dartmouth College, 2025.10
- Statistical Limits of LLM Alignment ICSA 2025, University of Connecticut, 2025.06
- Algorithmic Bias of RLHF Siam MDS 2024, Atlanta, 2024.10
- Algorithmic Bias of RLHF JSM 2024, Portland, 2024.08
- Adversarial Rademacher Complexity COLT 2024, University of Alberta, 2024.06
- Stability Analysis of Adversarial Training IOS 2024, Rice University, 2024.03
- Adversarially Robust Generalization IOS 2024, Rice University, 2024.03
- Algorithmic Bias of RLHF University of Pennsylvania, 2023.11
- Uniformly Stable Algorithms of Adversarial Training MOPTA 2023, Lehigh University, 2023.08
- Stability Analysis of Adversarial Training Sichuan University, 2023.02
- Adversarial Rademacher Complexity Sichuan University, 2023.02
- Stability Analysis of Adversarial Training AI Times Forum, 2022.10
- Metric Entropy RL Seminar, CUHK(SZ), 2020.08

Poster Presentation

- Restoring Calibration for Aligned Large Language Models ICML 2025, 2025.07
- Algorithmic Bias of RLHF 4th Penn Conference on Big Data, University of Pennsylvania, 2024.09
- Algorithmic Bias of RLHF ASSET Center AI Research Mixer, University of Pennsylvania, 2024.09
- Uniformly Stable Algorithms for Adversarial Training ICML 2024, 2024.07
- Algorithmic Bias of RLHF RL workshop, COLT 2024, University of Alberta, 2024.06
- Adversarially Robust Generalization NeurIPS 2023, New Orleans, 2023.12
- Adaptive Smoothness-Weighted Adversarial Training AdvML, ICML 2023, Hawaii, 2023.07
- Adversarially Robust Generalization AdvML, ICML 2023, Hawaii, 2023.07
- Uniformly Stable Algorithms for Adversarial Training ML Safety, NeurIPS 2022, New Orleans, 2022.12
- Stability Analysis of Adversarial Training NeurIPS 2022, New Orleans, 2022.12

AWARDS AND SCHOLARSHIPS

- NeurIPS 2024 FITML Best Paper Runner-up 2024
- NeurIPS 2023 Scholar Award 2023
- NeurIPS 2022 Spotlight 2022
- Tencent Ph.D. Scholarship 2020 - 2023
- Poster Award of 2021 Graduate Research Conference at CUHK(SZ) 2021
- Postgraduate Scholarship by The Chinese University of Hong Kong, Shenzhen 2017 - 2023
- Undergraduate Scholarship by Sun Yat-sen University 2013

TEACHING

Instructor

ICSA 2025, University of Connecticut

- Statistical Inference in Large Language Models

Summer 2025

Invited Guest Lecturer

University of Pennsylvania

- STAT 9917: Statistical Topics in Large Language Models

Spring 2025

Teaching Assistant

CUHK(SZ)

- MAT 3220: Optimization II
- MAT 3007: Optimization I
- MAT 4001: Numerical Analysis
- MAT 1004: Mathematical Analysis II
- MAT 1003: Mathematical Analysis I

Spring 2019 & Spring 2020

Fall 2019

Fall 2018

Spring 2018

Fall 2017

I conducted tutorial sessions in English for students twice per week per course, as CUHK(SZ) is a university that employs full English medium instruction.

PROFESSIONAL SERVICE

Session Organizer & Chair

- 2024 First Macau International Conference on Business Intelligence and Analytics
– Session: Optimization and Statistics for Large Language Models
- 37th Annual Conference on Learning Theory (COLT 2024)
– Session: Adversarial and Robust Learning
- Modeling and Optimization: Theory and Applications (MOPTA 2023)
– Session: Optimization Algorithms of Adversarial Training

University of Macau, 2024.12

University of Alberta, 2024.06

Lehigh University, 2023.08

Journal Reviewer

- Journal of Machine Learning Research (JMLR).
- IEEE Transactions on Information Theory (TIT).
- Journal of the American Statistical Association (JASA).
- Machine Learning.
- Quarterly Journal of Economics and Management (QJEM).

Conference Reviewer

- Machine Learning: NeurIPS 2022-2025, ICML 2023-2025, ICLR 2023-2026, AISTATS 2022 & 2026.
- Computer Vision: ECCV 2024, CVPR 2023, ICCV 2021.

MENTORING

- Zhanliang Wang (on Project [C1])
- Ruochen Jin (on Project [C2, C1, P2])
- Nikhil Ruia (on Project [P2])
- Zhekun Shi (on Project [J2, P3, P1])
- Kaizhao Liu (on Project [J2, P3, P1])
- Bohan Zhang (on a project of preference alignment)

Master student at UPenn

Master student at ECNU

Undergraduate student at UPenn

Undergraduate student at Peking University

Undergraduate student at Peking University

Undergraduate student at Peking University

LANGUAGES

- English: Fluent
- Chinese Mandarin and Cantonese: Native

INDUSTRY EXPERIENCE

GF Securities & Futures

Financial Engineer

Guangzhou, China

2015 - 2017

- Worked on option pricing and modeling. This experience helped clarify a strong long-term interest in theoretical research, which led to a transition to Ph.D. training.

REFEREES

Professor Weijie J. Su

Associate Professor

Department of Statistics and Data Science

The Wharton School

University of Pennsylvania

Address: 411 Academic Research Building,
265 South 37th Street, Philadelphia, PA 19104

✉ suw@wharton.upenn.edu

Professor Qi Long

Professor and Vice Chair

Department of Biostatistics, Epidemiology and Informatics

Perelman School of Medicine

University of Pennsylvania

Address: 201 Blockley Hall,
423 Guardian Drive, Philadelphia, PA 19104

✉ qlong@upenn.edu

Professor Zhi-Quan (Tom) Luo

X.Q. Deng Presidential Chair Professor

Vice President (Academic)

The Chinese University of Hong Kong, Shenzhen

Address: 620 Administration Building,
2001 Longxiang Boulevard,

Longgang District, Shenzhen 518172, China

✉ luozq@cuhk.edu.cn

Professor Ruoyu Sun

Associate Professor

School of Data Science

The Chinese University of Hong Kong, Shenzhen

Address: 412 Daoyuan Building,
2001 Longxiang Boulevard,

Longgang District, Shenzhen 518172, China

✉ sunruoyu@cuhk.edu.cn